

Education

Usman Institute of Technology (Affiliated with NED-UET)

Karachi, Pakistan

Bachelor of Computer Systems Engineering.

2018

GPA: 3.5 / 4.00

Thesis: Rotten Fruit Detection Using Machine Learning (PyTorch CNN + Arduino Automation)

Experience

Datathon - National Planning Council, Qatar & Microsoft

Doha, Qatar

Data Analyst & Team Lead

May 2022

- Performed descriptive, predictive, and prescriptive analytics in Power BI on NPC's Datasets.
- Applied time-series forecasting and machine learning models (LSTM, Random Forest, XGBoost) to assess trends and accident severity in Python and Power BI
- Designed geospatial visualizations to highlight high-risk zones and inform public safety decisions.
- Presented findings aligning with Qatar National Development Strategy 2024–2030, aimed at reducing traffic accident mortality from 7.8 to 4.9 per 100,000 people by 2030.
- Recognized for innovation and invited to present at NPC's Executive Open Day.

Pakistan Council of Scientific & Industrial Research

Karachi, Pakistan

Intern

Aug 2017- Sep 2017

- Investigated "Telehealth Model 2022 for Flood Calamity in Pakistan" using augmented reality for a decision support system.
- Collected and analyzed disease data, identified accessible treatments, and defined system database specifications.
- Explored potential benefits of AR integration to support remote healthcare.

Projects

Rotten Fruit Detection Using CNN + Automation

Aug 2017- Aug 2018

- Designed a custom Convolutional Neural Network (CNN) model in PyTorch with convolutional layers, ReLU activation, and cross-entropy loss function for accurate fruit classification.
- Integrated Arduino UNO with Python & IR sensors for real-time fruit sorting automation using the servo.h library.
- Preprocessed images using OpenCV to ensure clean data input for the model.

Smart Traffic Light Management System

Jan 2018- Feb 2018

- Developed a CNN model using Edge Impulse for real-time traffic signal management based on vehicle detection.
- Implemented dynamic signal timing and emergency scheduling mechanisms to optimize traffic flow and prioritize heavily congested lanes.
- Deployed the solution on Arduino Nano for efficient real-time processing and scheduling.

Job Market Analysis with Power BI

March 2022

- Analyzed job market trends in data science industry using Power BI for dynamic visualizations and trend analysis.
- Cleaned and processed data using Power Query; utilized DAX to create interactive dashboards highlighting key trends and skills in demand.
- Developed a Power BI dashboard for the DataSearch recruitment team to aid in identifying in-demand skills for data roles.

Segmentation Techniques for Rotten Fruit Detection

Jan 2018- Feb 2018

- Implemented three segmentation techniques from an IEEE research paper on MATLAB for detecting rot in fruits.
- Suggested two new methods for segmentation using different color models and filters.
- Performed a comparative analysis to assess the effectiveness of each method for fruit rot detection.

Rent a Car System (SQL Database Project)

Feb 2018-March 2018

- Designed a relational database system using SQL for a car rental service, incorporating tables for customers, bookings, vehicles, and insurance.
- Implemented business logic with primary/foreign key relationships, data integrity constraints, and triggers to enforce business rules such as minimum booking amounts and vehicle ID validation.

Skills

- **Programming Languages:** Python (Advanced), C++ (Basic), MATLAB (Intermediate)
- **Analytics & BI:** Power BI (Advanced), DAX (Intermediate), Excel (Intermediate), Matplotlib (Intermediate), Seaborn (basic), Pandas (intermediate), NumPy (intermediate), Databricks (Basic)
- **Databases & ETL:** SQL (Advanced), MySQL (Advanced), PostgreSQL (Intermediate), SQL Server (Intermediate), PySpark (Intermediate)
- **Machine Learning & AI:** Scikit-learn (Intermediate), PyTorch (Intermediate), TensorFlow/ Keras (Intermediate), Azure (Basic)
- **Image Processing & Vision:** OpenCV (Intermediate), Scikit-Image (Intermediate), Pillow (Basic), Edge Impulse (Intermediate)
- **Networking & Systems:** Packet Tracer (Beginner), OSI Model (Beginner), IPv4/IPv6 (Intermediate), Operating Systems and Networking concepts
- **IoT & Embedded Systems:** Arduino (Intermediate), Raspberry Pi (Intermediate)
- **Other Tools:** MS Office (Advanced), StarUML (Intermediate), Proteus (Intermediate), AutoCAD (Basic), Git (Basic)
- **Laboratory:** LaTeX (Intermediate, for research writing)

Certifications

- Generative AI Concepts – DataCamp
- Building Recommendation Engines with PySpark – DataCamp
- SQL Server for Database Administrators(SkillTrack) – DataCamp
- Computer Vision with Embedded Machine Learning – Coursera
- Image Processing with Keras in Python– DataCamp
- Pandas – Kaggle
- Intro to Machine Learning – Kaggle

Languages

- Urdu (Native)
- English (Fluent)